

PYTHON APPLICATION PROGRAMMING

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ASCII CODE

Char.	ASCII	Char.	ASCII	Char.	ASCII	Char.	ASCII
A	65	N	78	a	97	n	110
B	66	O	79	b	98	o	111
C	67	P	80	c	99	p	112
D	68	Q	81	d	100	q	113
E	69	R	82	e	101	r	114
F	70	S	83	f	102	s	115
G	71	T	84	g	103	t	116
H	72	U	85	h	104	u	117
I	73	V	86	i	105	v	118
J	74	W	87	j	106	w	119
K	75	X	88	k	107	x	120
L	76	Y	89	l	108	y	121
M	77	Z	90	m	109	z	122

BRANCHING

- Branching allows us to run different statements for different inputs.
- It is helpful to think of an **if statement** as a locked room, if the statement is **True** we can enter the room and your program will run some predefined tasks, but if the statement is **False** the program will ignore the task.

IF STATEMENT

```
# If statement example
```

```
age = 19
```

```
#age = 18
```

```
#expression that can be true or false
```

```
if age > 18:
```

```
#within an indent, we have the expression that is run if the condition is true
```

```
    print("you can enter" )
```

```
#The statements after the if statement will run regardless if the condition is true or false
```

```
print("move on")
```

```
you can enter
```

```
move on
```

IF – ELSE STATEMENT

```
# Else statement example
```

```
age = 18
```

```
# age = 19
```

```
if age > 18:
```

```
    print("you can enter" )
```

```
else:
```

```
    print("go see Meat Loaf" )
```

```
print("move on")
```

```
go see Meat Loaf
```

```
move on
```

ELIF STATEMENT

- The elif statement, short for else if, allows us to check additional conditions if the condition statements before it are False.
- If the condition for the elif statement is True, the alternate expressions will be run.

```
: # Elif statment example

age = 18

if age > 18:
    print("you can enter" )
elif age == 18:
    print("go see Pink Floyd")
else:
    print("go see Meat Loaf" )

print("move on")

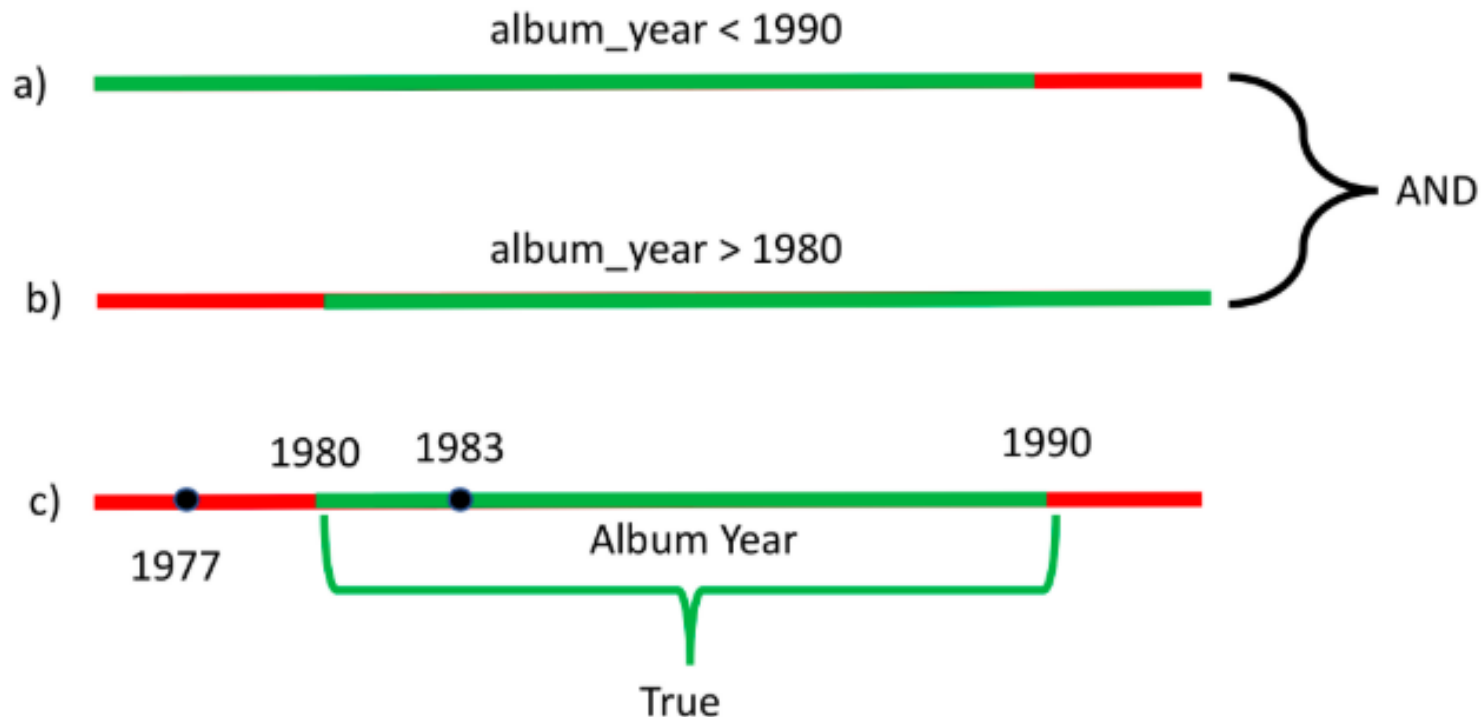
go see Pink Floyd
move on
```

LOGICAL OPERATORS

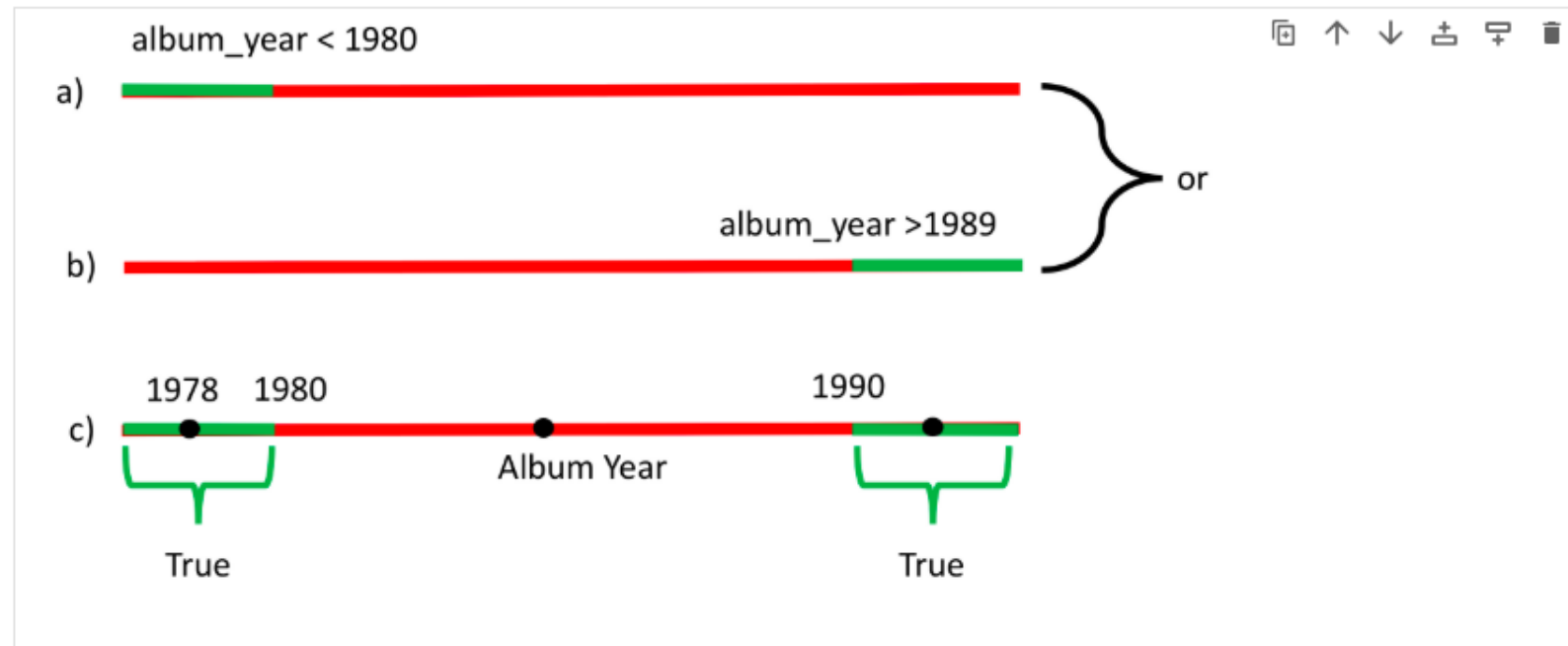
A	B	A & B
False	False	False
False	True	False
True	False	False
True	True	True

A	B	A or B
False	False	False
False	True	True
True	False	True
True	True	True

- Determine if an album was released after 1979 (1979 is not included) and before 1990 (1990 is not included). The time periods between 1980 and 1989 satisfy this condition. This is demonstrated in the figure below.
- The green on lines **a** and **b** represents periods where the statement is **True**. The green on line **c** represents where both conditions are **True**, this corresponds to where the green regions overlap.



- Determine if an album was released before 1980 (1979 and earlier) or after 1989 (1990 and onward), an or statement can be used.
- Periods before 1980 (1979 and earlier) or after 1989 (1990 and onward) satisfy this condition. This is demonstrated in the following figure, the color green in **a** and **b** represents periods where the statement is true. The color green in **c** represents where at least one of the conditions are true.



Range

- Range object as an ordered list.
- For now, let's look at the simplest case. If we would like to generate an object that contains elements ordered from 0 to 2 we simply use the following command:

```
range(3)
```

FOR LOOPS

- The for loop enables you to execute a code block multiple times.

```
: # For Loop example  
  
dates = [1982, 1980, 1973, 1996]  
N = len(dates)  
  
for i in range(N):  
    print(dates[i])
```

```
1982  
1980  
1973  
1996
```

The code in the indent is executed N times, each time the value of i is increased by 1 for every execution.